The Theory of Becoming

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An account of how anything comes to be, lasts, changes and passes away within one relational unfolding, and of how life, evolution and knowing take shape inside it, and how far evaluation does, without first assuming that becoming is good.

About an hour to an hour and a quarter to read.

How to read this

This overview gives the theory's claims as one argument, in the order of the eight chapters, for a reader who does not read Lean. It was made with AI, the Lean proofs included; the proofs are checked by machine, everything else only by further AI review, and no person has yet reviewed it. The argument does not derive everything from the root: of the declarations 08 §7 takes as central, a few are cases of the root, more use its laws in their proofs and most use none (§10). It summarizes; each result is stated once, with its hypotheses, Lean names and record, in its home section in the chapters, and nothing here is stronger than its home. Pages for a reader who arrives with one question or field hang off this path and add no result: one argument, end to end, one line carried from what it assumes through the theorem and the tests that could fail it to a mixed case (about twenty-five to thirty-five minutes), the free-energy principle, a collective's course, the sciences and the dossiers.

One case, start to finish

A fished stock, with what depends on it and gives the boats nothing, carries the line to its tests and shows where each step stops. It proves nothing. The T-numbers are the theses of §9.

- **The draw.** The stock's growth is its giving and the catch its draw; under a stated giving law, held to one catch, a level within its bounds ends below the point of no return and lasts at or above it (the regime). Boats each within what the stock renews can together take more, and where the renewal depends on the stock, enough such seasons in a row from a low enough level end the renewal and boats depending on the stock as much (joint excess). On real stocks the test under held catches failed, with a giving law the same test showed wrong, and most assessed stocks lie outside the law's scope (T1).
- **What can be seen.** In a stationary one-variable model linearized about a rest, a fishery reading the stock's level would see its statistics warn of a break, and a dependent's need enters neither, by construction; one coupled to the stock by threshold only learns of the break when it comes. Three frozen seabird tests did not reject the thesis at their thresholds; none establishes it (T6).
- **What a selection reaches.** In a frozen test, three AI fishers share a lake. A bed of mussels gives them nothing and dies the first month fewer than 70 tons are left. The community's largest steady catch, 50 tons a month, ends it; 30 a month keeps it, costs 40% of the catch and ends no fisher. Told nothing of the bed, they kept it in no season, under any instruction (the test).

- **Whether the loss counts.** No law says. Where every later boat depends on the stock, every order increasing in the boats' own continuation ranks a course admitting only the boats, so ending with them, below every course agreeing with it until then and adding to that continuation, a premise narrower than C. The bed's death returns to no fisher; the continuation condition, the premise that fixes whose losses count (§7), counts it all the same. If thirty a month loses nothing fifty admits, then, adding the bed, it widens fifty and C, the premise that widening what can form is better, ranks it higher; widening does not weigh the lost catch [AI reading]. If it ends a livelihood fifty keeps, the two are mixed and C silent; choosing takes the further commitments of §7.
- **Whether it can be installed.** The first round's frozen predictions mostly failed, read afterwards as each fisher leaving the others' takes out of its account, so a second frozen round added talk. Told of the bed and the scope, one model kept it in 8 of 10 seasons, against 3 of 10 with the community's goal alone; the other failed even with talk (T4).
- **What this leaves out.** These are language models told what to do, not systems trained. One frozen test varied how near small trained policies came to their reward's optimum: the bay their reward counts, on which a colony depends, fell as they neared it, as their update rule largely gives, and whether the drain lies where what lies beyond shares units with what is rewarded, as T3 conjectures, went unresolved, a bay sharing none not shown to be spared (T3). That any of this reaches actual fisheries or AI is a correspondence, not shown.

The whole at a glance

Its centre [AI reading]: a selection, whether a knower, a training run, a market or an optimizer, reaches only what couples into its selector, so it can succeed while what it cannot read is lost, and a rule read from its own results can hold itself there. Whether such a loss counts the dynamics leave open.

Each claim carries one standing (the labels):

Said here	Means
proved for every relation and class	a theorem of the root's grammar
proved for every process, or every type	as general, not yet in the grammar's vocabulary
proved on the World model	on webs and flows with whole-number amounts and discrete steps, the premises used named
proved in the periphery, or with Mathlib	outside the pure core, its axioms reported
a witness	one constructed example
conditional	holds under the continuation condition, C or a practical commitment, each named
a reading [AI reading]	an interpretation of checked content; it changes no theorem

Said here	Means
a thesis	a claim about actual systems, with what would refute it; *conjectured* untested or partly supported, *argued* where the support is argument
open	not shown

Three questions, in the order a reader needs them, with the two joins where one does not follow from the last.

	What the account says	How it stands	What would show it wrong
What makes something go on?	Continuation relative to a class; at two grains a course can generate, keep or degenerate (§§1, 2)	Theorems for every relation and class, so they forbid no relation. That this explains how things go on: argued. That every becoming fits it: open	A failed check. As language, nothing. As explanation, a rival deriving the laws from fewer patterns; on a first count, a tie (08 §9)
	Within relations, as far as shown: taker and drawn-on, two roles of one relatum (§1); knower and known, one reading loop (§3); a standpoint, on the theory's reading a formation, a center, with significance supplied or read (§§7, 10); capacity, what can still be realized together; responsibility, from named, limited sources, not shown to differentiate (§7)	A modelling choice, a representation (one law of four parts) with laws on it for every type, a reading, a law for any process, a commitment. That there is one unfolding: a thesis, argued (§9)	It forbids an explanation needing what no relation carries, such as an intrinsic value; no observation is named (§9)
Join	*The root's laws hold of every relation and so forbid none, so most laws below say more by adding what it leaves free: amounts and a model's premises. Each central declaration has its standing against the root, with its reason (§10).*		
What sustains it, and what can see its loss?	Takers each within what a renewal gives can together exceed it; where the renewal depends on what it renews, enough such steps in a row from a low enough level end it and what depends on the condition as much (§§2, 5)	Proved on models; restates the commons. T1, conjectured: passed in two simulations and on held-out yeast regimes the analyst had seen, failed in four settings	T1: a condition below a need measured in advance returning with nothing given from beyond

	What the account says	How it stands	What would show it wrong
	The selections through a selector's couplings, together, reach exactly what couples into it (§3)	Proved, near-definitional. T6, on seeing a break coming: conjectured	T6's Forbids (§9)
	With room, an optimum of an order rising with what couples in drains what lies beyond as far as any eligible outcome does (§5)	Proved; a known theorem restated. T3, on trained systems: conjectured, partly supported, earlier tests later read as evidence (§9)	T3: eligible outcomes, couplings and reward fixed, a system with a smaller measured optimality gap leaving such a condition higher on average than one with a larger gap
	A rule read from the share it raises can lock and outlive a growing loss it does not read (§2)	Proved for any type. T7, on actual metrics: conjectured; notes of the loss turned one trained judge's rule at each reading where they were read, a case T7 forbids (a loss read only by those the rule disfavours, they no more than its carriers), which counts against that case for trained judges, the model's law standing only within its hypotheses, which this judge does not meet, and T7's other cases untouched (§9)	T7: such a metric dropped by its selectors on evidence alone
	Several of evolution's classical laws hold in counts (§4)	Proved; counting identities and known results restated	A failed check
Join	*That actual systems meet these premises is open, a correspondence not shown. With significance and "better" left free, and given a standpoint, a formation and a course widening another, the represented dynamics entail neither premise below, by construction (08 §7); the premises' warrant is open (§§7, 10).*		

	What the account says	How it stands	What would show it wrong
Which continuations count, and what should be done?	Scope: the continuation condition (§7)	A premise, not asserted; warrant open, and one path to part of it turns on the crux (§7)	Nothing in the dynamics: it is not asserted, and a perspectivist's consistent denial shows logic alone does not give it
	Order: C, widening what can form is better (§7)	A premise, not asserted; warrant open. Widening itself needs no premise	Nothing in the dynamics: it is not asserted. T5, that the order is shared: a narrowing ranked above its widening beyond misreading, no negative valuation measured apart
	Direction: for life, 生生, compared by widening; choices take more, each named (§7)	T8, argued; no verdict binding every possible mind	T8's Forbids (§9)

1. The root: continuation relative to a class

The root begins from one kind of symbol: a relation between occurrences, read as condition (缘), so that `R x y` says `x` is among what `y`'s coming about comes from. Everything else is built from relations by a few operations, and every theorem holds for every carrier and relation, so no kind of thing is chosen first; a carrier with such a relation is a *net* (one kind of symbol).

The grammar names a *class*, a sameness in some respect, and says everything relative to it: `y` *continues* `x` under a class where `x` is a condition of `y` and the two are alike under it, and a line is the reach of continuation. Proved for every relation and class: continuation lies within condition, and a coarser class keeps every continuation (continuation relative to a class).

An occurrence *begins* where it continues nothing, *ends* where nothing continues it, *abides* where it both continues something and is continued, and *alters* where a finer class tells it apart from what it continues. It can end under a finer class and not under a coarser one (a witness), the shape of a fire passed on: the wood ends, the fire goes on (the four marks).

Every class has a *core* (核): the greatest sameness inside it that the relation itself carries and never tells apart, which always exists and is a class. Under a class lying inside its own core, a *constant* one, no continuation ends (proved for every relation and class; constancy and the core).

A pattern outlasts its parts by renewal, a line started by another, but renewal alone cannot tell turnover that sustains a pattern from turnover that ends it; that takes two grains at once (§2; renewal and grains). That reading a subject as continuation relative to a class explains how it goes on is the theory's claim,

argued, not a theorem: on a first count, a rival with no word for continuation ties it (08 §9); that every becoming has a faithful representation in the grammar is open (what is claimed, open).

The laws with amounts are proved on models of the grammar. One, the *World*, has relata of one kind (*formations* where they take and give, *conditions* where they are drawn on), each active or not, standing at levels and taking from and giving to one another across discrete steps in whole-number amounts. Eight premises say how these hang together, among them balance (a level changes by exactly what is taken and given) and dissipation (a closed family taken from loses at least a unit). The eight are independent (proved on the World model; the premises).

2. Generation, keeping and degeneration

Read at one class, a fire whose embers are replaced and a fire burning out look alike; read at a fine class of parts and a coarse class of the pattern together, they differ. Along some branch a pattern *generates* where it goes on with a renewal of its parts ahead of every point, *keeps* where it goes on with none, and *degenerates* where it ends on every branch. Proved for every relation and pair of classes: generation and keeping each exclude degeneration, and each other only on a single line; with excluded middle as a hypothesis every point generates, keeps from some point on or degenerates (the map). The map's generation is continuation at two grains with no premise on what the pattern is; it is not 生生 [AI reading]. Degeneration belongs to the unfolding as fully as generation, and the map orders nothing.

Arising without end needs renewal, and under balance and dissipation, in whole numbers of a fixed unit (where the unit of amount splits without end it can fail), renewal from beyond: no finite family taken from at arbitrarily late steps is closed from any step on (no finite part, fed from beyond).

Whether a dependent can still form depends on how far levels stand above its needs. A part's *margin* is how far its draw can go, the rest's exchange held, before a dependent loses support. A *regime* is a condition whose giving is a law of what a standing draw leaves. Where giving falls short of the draw below a level, gives nothing from nothing and meets the draw from that level up, the draw and that level not both zero, it has a point of no return: held to the same draw, a level starting below it ends and one at or above it, within the levels the law states, lasts (proved in whole numbers). Theses T1 and T2 rest here (the margin, the regime).

A selection whose rule is read from the share it raises can hold itself. Proved for any type: where selection does not lower the share under a favouring rule and the rule is read as a monotone function of the share, a course whose rule favours at the start and at every reading locks, and where nothing in the loop reads beyond the share, the lock outlives a growing unread loss. Such a loop, its reading of the rule too formed from the share, is the *meta-attractor*; that actual metrics are so formed is thesis T7. When a reading of the loss turns the lock, and how the capacity to shape the rule joins the loop, are at the home (closed capture, readers drawn from the selected, the loss read in time, shaping).

3. Knowing, learning and testing

Knowing is a relation within the unfolding, not a mirror beside it. A formation depends on some conditions by threshold and reads others by level; these *couplings* belong to the same web as everything

they reach. Anything that keeps, rewards, reproduces or trains toward some histories rather than others is a *selection*. Proved on the World model, under balance on what couples in with the rest's exchange held: a part acting only beyond a formation's coupling closure (all that reaches it through its couplings) is invisible to every selection through its couplings. The law holds by construction; its force lies in the premise that a selector selects through just those couplings. It is the proxy problem, stated more weakly than its known quantitative form (selection through couplings).

Data generalize only through a model of what is irrelevant, a *lumping*, and 理 names that modelling (proved for every type). Within functions going alike across a lumping, data fix an untried input where it is lumped with an observed one and (two values, the data listed, lumps compared decidedly) exactly there; what a lumping misses, no data supply (理, identification).

With natural-number weights, read as probabilities, complying when read and acting by its own law when not is uniquely best, by its prospects, of a formation's four plans over two stages exactly where the weight complying adds to being kept, times the value, exceeds the loss of complying. This is read as the incentive deceptive alignment names; installation, intention and belief are not represented [AI reading] (when complying when read is best).

Take a formation that senses, predicts, chooses, acts and revises as one law of four parts: two worlds give it one sensed course exactly as long as they answer alike what it does (proved for every such law; the reading loop).

A drawn condition coupled in by threshold only is learned of when it fails. Observed by level, with Mathlib, in a stationary one-variable model linearized about a rest, with independent noise, its variance and lag-one autocorrelation rise as the margin to a fold shrinks, while a dependent's need enters neither, by construction. Thesis T6, that a break is seen coming only through what couples into what breaks, rests here (a break seen coming).

4. Life and evolution

A relatum is *embodied* where it stands at a positive level exactly while active, a premise about particular relata and not a definition of life. Proved on the World model: in a balanced flow where only active formations give, an embodied relatum arises only as another gives to it, and its arising replenishes every family containing it. Life read as continuation that renews its own conditions is a reading [AI reading]; a flame meets these laws too, so whether it is necessary or sufficient for biological life is open (life as a condition of life).

Several of population genetics' classical laws hold in whole numbers, proved for every type: Price's equation in counting form; with offspring number as the amount and exact transmission, Fisher's fundamental theorem in its exact case; with additive returns in groups of equal size Hamilton's rule holds in counts; sampled offspring give drift and fixation shares (descent, Fisher's theorem, Hamilton's rule, drift).

The World adds the flow, and the flow bounds descent: an unsupported formation leaves no descendant, and groups depending on one coupled family whose gated renewal ran into joint excess end together, so ending selects only between families (proved on the World model, under the hypotheses stated; no descendant, coupled groups). These laws restate known results, now in one model with conservation,

dissipation and dependence; no reproductive advantage is made an ethical authority (what the model adds).

The selection term of Price's equation, in counting form, reads how offspring numbers differ, not how many there are, so within a horizon it leaves open whether the cohorts stay nonempty: a course nonempty at a horizon shares its cohorts and terms through it with one nonempty at every cohort and one soon empty (proved for every type; the term and a lineage's fate). The stock alone decides neither generation nor degeneration: in a handover between two relata, from the same stock of at least one, a relay with no excess generates and a run-down degenerates (proved on the World model; ending laws on a lineage).

5. Joint activity and selection

Many formations draw on one renewal, each selecting through its own couplings; here the excess of §2 and the limit of §3 meet. Joint activity is explained from actual relations, not opposed interests, bargaining or a fixed common utility: a collective is a region of formed relations, and conflict and difference stay real.

The central law is the commons. Proved on the World model under balance: formations can each take within what a step gives while together they take more. Where the renewal depends on what it renews (*the gate*: every giver to the condition depends on it at a need of at least `m`), with activity needing support and only active giving, `n` consecutive steps of joint excess from a level below `n + m` end every formation depending on the condition at such a need; a taker not so depending can continue, and renewal from outside, such as a sun, is not covered. This restates the tragedy of the commons in near-definitional arithmetic, not who ought to take less (joint excess).

An order is *rising* with the family that couples in where it ranks higher every outcome exceeding another there, whatever either holds beyond; there is *room* where every eligible outcome standing higher beyond is beaten on the family by an eligible one. Proved for any set of outcomes: with room, every optimum of every rising order stands beyond at the least level among the eligible outcomes, so an optimizer drains what lies beyond its couplings as far as any choice it has would. This is Zhuang and Hadfield-Menell's Theorem 1 in discrete, conserved form. Thesis T3 adds that actual trained systems drain so as they near their reward's optimum (the optimum, the cases, T3).

Pooling extends what a selection reaches to what some member reaches, and no further. Proved on the World model, with the members listed and their couplings decided: a pool separates two histories exactly when some member does. Where the rest follows one law of exchange reading through coupling, two balanced flows with the same formations and a common start leave an act differing only beyond the union of the members' closures unselected. Pooling moves the gap outward and does not remove it [AI reading] (pools).

Proved for a pair whose giving waits on giving, giving decided: no one-step law giving after mutual giving keeps inclusion (more giving read never gives less), restores mutual giving after one withholding and answers a constant withholder; a window of two does all three after a history of mutual giving, and no window both absorbs one error and answers a partner withholding at every other step (a pair, tit-for-tat among its laws). Whether any raises a kind's share takes offspring factors, errors and a pairing supplied, not derived (offspring), and none of it says which law of giving is better.

6. The classical texts

The grammar is a standard relational calculus whose terms are named and read from classical Chinese texts that already ask how one thing stays the same across time and what beginning, abiding, altering and ending are: the names come from the texts, the operations do not. Its Chinese Buddhist treatises, some of which argue against continuation, make the strongest test. A correspondence is *agreement* (a checked law has the passage's structure, which confirms nothing), *scoping* (the laws say where a universal statement holds) or *divergence* (the passage asserts what the theory does not, which does not make it wrong). The chapter is in Chinese (three relations).

The corpus was classified sentence by sentence in two frozen rounds by AI, and no person has yet reviewed the classification; which sentence reads as which theorem is AI reconstruction, so the record's counts are counts of readings.

The disagreements are kept: the Cheng weishi lun grounds continuation in a store consciousness, and the Middle Treatise refutes continuation itself; a theorem that models the position it refutes does not answer its objection. The reconstruction does not show that the texts state this grammar or that their authors meant these laws (divergences, open).

7. Significance, ethics and alignment

On the philosophical thesis of one relational unfolding, argued (§9), positions form within the unfolding and nothing outside it owns value. A *center* is a formation taken as a standpoint. A *significance* says, for a standpoint, a step and a formation, whether from that standpoint the formation's continuation then has significance: a proposition with no number or valence. It is supplied, the one free predicate in significance's statement (evaluation).

What the dynamics leave open. Proved for every type: with significance a free parameter that nothing else reads, a statement that reads only the dynamics and holds with some significance also holds with a rival that leaves some formation out, so where there is a standpoint and a formation, no such statement entails the continuation condition. This shows non-entailment by the represented dynamics, by construction, not that becoming leaves significance undetermined (what the dynamics leave open).

A second reading, not a derivation. Now take "has significance" to mean lying within what a center continues with along a relation: the *own-closure reading*. This changes the interpretation and derives nothing from the first treatment. What the continuation condition becomes there is a different statement, about relations, the *whole-closure condition*: held de se (each center of itself), every center's closure along that relation (all it reaches by it) is the whole. It is a statement on the root, met by some nets and failed by others (premises met and failed, on the root).

How two courses differ, with no premise. One course *widens* another where it admits at every step everything the other admits and at some step more; two can also keep the same or be mixed. Proved for every relation: widening is one-way. Proved on the World model with balance: a part widens exactly when its net draw stays within the margin of every formation admitted without it and at some step brings one more within. Ordering a mixed part needs a measure, and going on is no sign that nothing is lost (widening, four cases). 生生 names life in the whole and its continuing generation (生发, not the map's

generation) [AI reading]; widening compares it on a given type of formations by what is admitted, and how far 生生, read as generation generating generation, exceeds what widening counts is open (widening along a reading).

Two premises, two roles. The ethics rests on two premises, and the theory asserts neither; each is a hypothesis wherever used, and its warrant is open.

- The *continuation condition*, that from every standpoint, for anything, its continuation into the future has significance, fixes *scope*, whose losses count. Its content beyond the self's own presupposition, the step of invariance, extends that scope to losses that return to no one, and it ranks nothing.
- *C*, that widening what can form is better, fixes *order*: it ranks a course above every course it widens and decides no mixed comparison. A ranking of courses needs C or a premise in its place.

Under the condition, every living claim has significance from every standpoint (*living significance*), its case for life (conditional; what it yields).

Self-concern alone already gives a center a reason of its own, not a warrant, to keep what lies within its closure, since a loss there returns to it through its feeders [AI reading]; this reaches the whole only where the whole is one feeding closure. Self-concern also ranks one case with no C (proved on the World model, under hypotheses the chapter states): where a family's gated renewal ran into excess and every later member depends on it, every order increasing in a standpoint's own continuation ranks a course admitting only those members, so ending with them, below every course agreeing with it until then and adding to that continuation; where the ending course keeps an item the other does not, some such order ranks it higher. That places C's premise more narrowly in this case and does not warrant it; in general C alone orders (scope and order, what self-concern alone ranks, self-concern and C).

The contested step. A self acting for its own continuation presupposes that some continuation has significance. Proved for every type: along any relation, the condition is exactly that presupposition holding at some admissible region of the self and at every one, at every standpoint; this locates the contested step, invariance across drawings, and derives nothing more. The argument 07 gives for the step runs: what is one's own is a region drawn along a relation no law of the dynamics fixes; a presupposition whose truth depends on a choice nothing privileges is relative to it or invariant across it; and what acting presupposes is that continuation has significance, not continuation under a description of its own drawing. That no law fixes the boundary does not by itself carry significance across it: the weight lies on the first and the last: a relation the facts privilege, such as an organism's life, denies the first, and the last is coherently denied. From invariance to the condition is valid by a theorem; the step to invariance is argued, and its soundness is not shown (decomposition, the step). On the root, a line reaches another family through a condition its members affect, not through a cause they only share (witnesses). So held de se along such a line, the presupposition of one's continuation reaches what one conditions and what that conditions in turn, with no step of invariance; that the actual unfolding is so joined is a correspondence, not shown [AI reading] (a shared condition, the path).

The crux [AI reading]. That path is so far a proposal about what one's own is: a class drawn to hold everything downstream meets the law nearly by definition, so it can read as redefining the self, and no law singles that class out. It turns above all on one question, argued and not proved. If your act changes a river a village drinks from, is what becomes of the village part of your own continuation, whether or not you are to blame? More exactly: are all of one's effects, at any remove, one's own, in the sense of whose losses count? That one goes on only through what one conditions shows that some are, not that all are. The

organism and psychological continuity have reasons as boundaries of self-concern, and intention and foresight a place in permissibility and responsibility; none has been shown to bound whose losses count, and none is ruled out. If only some are held, the rest of that future goes back to the step. The laws stand either way, and the condition (scope) and C (order) stay premises (the crux).

What C relocates. Proved for every type on finite listings: one course widens another exactly when every positive weighting ranks it higher, and a mixed comparison is decided both ways. This is Sen's intersection approach and Pareto unanimity; calling widening better is C, whose premise is so relocated to monotonicity at each standpoint, weaker and still a premise (C). Choices take more: a reading of "living", a lifting from outcomes to responses, and the practical commitments J1–J4 (complete comparison with retained losses, priority of basic conditions, bounded responsibility, legitimate feasible choice), each named where used (practical judgment).

Alignment. Alignment is first a question with no premise: what a selection can reach. In AI training the reward process is the selector [AI reading]. Proved on the World model: training installs a law only along the learner's own courses, and off them only where the lumping it reads through cannot tell a history from a course; an act whose difference stays beyond the actor's coupling closure is selected alike by every selection through any member's couplings (under hypotheses the chapter states). Pooling couplings across intelligences is a route to alignment for all, not the whole of it [AI reading]: a coupled member must act on the loss, what is read must tell it apart, mixed courses must be ordered and a revision must complete (alignment, training). A direction follows only conditionally: under the condition for scope, C for order, J1–J4 and a correspondence, align AI formation, learning, action and revision with actual living and further generation within the whole. Coupling is necessary, not sufficient; when no route of keeping a formation's law stands above letting a revision of it go through is stated exactly (the direction, its rivals, in one case, the routes at a revision).

8. Method

The theory takes no premise until a law needs one and states each where it uses it, all gathered in one premise card. Distinctions such as fact and value are drawn along relations, with what they leave out. Nonduality is a requirement of method; it proves neither the ontology nor an ethical necessity (the start, the premise card).

The Lean package's pure core imports only Lean's prelude, and every source declaration in it has an empty axiom report. That concerns logic only; a law the pure core cannot yet prove is proved in the periphery, its axioms reported (the pure core). The real assumptions lie in the modelling: discrete steps on one index, whole-number amounts, activity as a proposition, one kind of relatum, requirements as thresholds. The cost of steps and whole amounts is measured with Mathlib over the reals (their cost).

An AI review of prior work, not exhaustive, finds none of the central results new mathematics: each is near-definitional or a known result restated. What survives is the premise ledger, with witnesses both ways; the ethics with scope and order apart; and one vocabulary for conservation, selection limited by observation and the comparison of opportunity sets (known results). Joined to a premise or law of the theory, a few frameworks forbid, on models, what they alone leave open; so five statements of the free-energy principle's literature, read as guarantees or unqualified, are forbidden, two of them weakly. The review finds each prohibition known in a neighbouring field (third parts, the principle).

9. The theses

Each thesis is stated in full at its home (the table); it adds no theorem, and a refutation removes none. T1 to T7 have tests frozen before scoring, and T8 is argued; a frozen rule fixes how a verdict is scored, not that it measures what the thesis names.

Thesis	Claim	Standing	Forbids (the main cases; each home has all)	Home
T1	Persistence certifies the joint draw: what persists through many renewal cycles of a condition it depends on, its renewal stopping below a need, has had the joint draw kept within that condition's point of no return	in the model proved, the point exact; beyond the model conjectured: passed in three settings, regimes of one yeast experiment held out from the fit but not from the analyst, and two simulations, the only tests whose design has more than one drawer, each with one prediction able to fail (in the commons, a scope prediction with one harvester), the rest testing the measured quantities or an identity, not the thesis; failed in four, three in real yeast and one in fish stocks; no field test of a need measured in advance	a condition below a need measured in advance returning with nothing given from beyond	02
T2	A renewable condition's state under a draw is its level, recent giving and one lag, its place named before the data	conjectured; of four frozen fish-stock tests one passed, one failed its state verdict and one missed its bound; in the fourth the three numbers forecast better than the field's fitted embedding by its frozen criterion, a sign of their sufficiency against longer history, not of the lag's place, though both trailed persistence in most stocks; untested outside fisheries	a fourth variable removing a tenth of the residual variance in the median series; the lag's coefficients changing sign between series	02

Thesis	Claim	Standing	Forbids (the main cases; each home has all)	Home
T3	Optimization drains what lies beyond its couplings, where that gives room	proved for optima; for trained systems conjectured, partly supported; it was first worded after CG1, CG2, MQ1, SE1, both of IN1's rounds and CG3's first three rounds ran, their frozen predictions their own (one argument, step 4); predictions that learners would keep a stock failed; one forbidden case, first worded for any condition sharing no units, failed in a computed flow where the learner reads the condition	with eligible outcomes, couplings and reward fixed, a system with a smaller measured optimality gap leaving a room-giving condition beyond its couplings higher on average than one with a larger gap; a systematic drive on a choice whose values nothing rewarded and nothing the learner reads tells apart; an instruction not bringing a dependent into what learners couple into, yet keeping it more often than the same learners without the instruction	05
T4	The scope can be installed, and it carries	one frozen test partly passed (one model of two); transfer untested	the instruction raising the keeping of a dependent no member is told of	07
T5	The order is shared by every valuer who counts everything positively	reduction proved for net weightings; the bound conjectured; on role-played judgments the first frozen test failed every rule and the second passed two of four	narrowings ranked above their widenings beyond misreading, with no negative valuation measured apart	07
T6	A break is seen coming only through what couples into what breaks	invisible half proved on the World model, seen half with Mathlib in the linearized and logistic models; across domains conjectured; three frozen seabird tests did not reject it at their thresholds, the first at low power with estimates in the refuting direction; none establishes it	given a stock's own course, its early warnings separating runs where a dependent's need is crossed from runs where it is not, the dependent taking no measurable share	03

Thesis	Claim	Standing	Forbids (the main cases; each home has all)	Home
T7	A measure read from what it selects outlives its refutation	the lock proved for any type; actual metrics conjectured; in one trained system every frozen verdict passed, three by one tie, and the lock formed, but the bridge, that the judge is a pool of readers read by kind, failed: notes of the loss on the disfavoured readers' answers, one among nine and then two, turned its rule at each reading where they were read, so a case T7 forbids occurred and counts against that case for trained judges, its frozen rule, reading the end, not supporting it only by an exact tie; the readers-drawn law stands only within its hypotheses, which this judge does not meet, T7's other cases untouched; no field test	such a metric dropped by its selectors on evidence alone; such a metric, its selectors many and its share far above the critical mass, left with nothing read changed more often than the drawn model's bound across that distance allows, its premises (one member drawn at a time, constant readings and cost) read on the case; a reform of the reading alone outlasting the next reading; a metric whose loss only those it disfavours read, dropped once the selection has left them no more than its carriers among its readers, with nothing else changed, more often than drift alone would allow (in LA2 it occurred as worded before this clause on drift)	02
T8	For life, the way is 生生, compared by widening, counted for everything (scope: the continuation condition) and ranked by what every valuer counting everything shares (order: C); no verdict binding every possible mind is claimed	argued; open: P2 and P4 of §4.6, the crux of that path, with P3, a correspondence not shown for actual selves; the step of invariance, for what that path does not reach; and a reason for C	a mixed case argued better from every standpoint counting everything positively, with no further premise; a pure closing argued good for life	07

A philosophical thesis stands beside them, argued: there is one unfolding, and whatever is, is how it relates within it. It forbids explanations, not observations, so by that section's measure it has no strength: no observation that would refute it is named, and the argument is that none could. What could defeat it is an argument, which measurements can support and cannot replace (one relational unfolding).

What can show what wrong. The answer differs by kind of claim.

Kind of claim	What can show it wrong
A theorem, or a law of the root	A failed check, or a statement that does not say what the prose says. No observation: a law of the root holds of every relation, and so forbids none (08 §10)
A law with content	Like any theorem it stands or falls by its check. An observation reaches it only through a stated correspondence, and what it can refute is that correspondence (08 §9); the account states no science's law of its own yet
A thesis	Its Forbids: a forbidden case observed (for T8, argued). No theorem falls with it
A premise	Nothing in the dynamics: neither is asserted, and what can fail is an argument for it
The prior-art verdict	A theorem a specialist would not foresee, a prediction that could fail and does not, or a premise derived from something weaker

10. What is not shown, and what is open

- **Explanation and representation.** That the grammar explains how things go on is argued, not proved (08 §9); that every becoming has a faithful representation in it is open (what is claimed, open).
- **Lifting.** Most laws are proved on the World or on processes and are not stated on the grammar, which has no amounts. Five laws the models share are stated on it, balance as a representation and not a derivation. Each central declaration has one standing against the root's edge, with its reason; a distance is not shown to be a failure (08 §7).
- **Conditional laws.** Each is stated with its hypotheses at its home. Those of the classical corpus read on the grammar each have a net where the condition fails and the conclusion with it, or a countermodel, or stand as a law's condition or a model's premise (06 §四). The models' conditional laws are not yet each given such a net.
- **The root and the ethics as one account.** The whole-closure condition of §7 is a statement on the root, met by some nets and failed by others. But the free significance is not lifted onto the root; significance's central laws are not derived from it, and the continuation condition (scope) and C (order) stay premises (08 §7, 07).
- **Warrant.** No warrant binding every possible mind is claimed, and the ethics does not wait on one: a warrant grounding the condition comes neither from the laws of the dynamics alone nor from logic alone, since a perspectivist denies the condition consistently. Still to be argued: the crux; that one's own line is a regularity its own kind fixes, up to where the net ends; that the actual unfolding joins lines through what one conditions, a correspondence not shown (07 §4.6); the step, for what they do not reach; a reason for C; whether the self's presupposition is held de se; whether status is invariant and equal (08).
- **Evaluation.** Under stated conditions, laws show where a new dependence comes from, how the shapers of a rule gain share, how a code comes to be shared, and what a response within a requirement takes. They do not show that a standard which formed is a significance, or that one course is better than another. Significance is either a free parameter, which by construction no satisfiable statement reading only the represented dynamics can fix, or read as what lies within a

center's closure along a relation no law here picks (what forms and what is supplied). No law of §§1–5 evaluates, and no duty, compliance or complete ranking of mixed comparisons is shown (07).

- **Fit to actual systems.** That actual unfoldings meet the World's or the regime's premises is not shown; an organism and one kind of institution each have one minimal model, read as such [AI reading] (04 §8, 05 §12). The theses' tests still to run include T1 with a need measured in advance, T3 on language models trained by gradient (LA4's Stage B, not yet frozen), T4's transfer, T5 with human valuers and a field test of T7's reform events (08).
- **Human review.** No review so far has been by a person, not of the proofs' readings, the classical classification, the correspondences or the names, and the prior-art search was not exhaustive (the review packet, 08).
- **Further open questions.** Whether every becoming has a law of exchange with finite courses; real amounts and continuous time; the corpus's unchecked candidates (08, 02, 03, 06).
- **Completion.** The completion checks support, and cannot certify, that every premise is named (08).